Probabilistic Reasoning

Professor Nihad Karim Chowdhury
Department of Computer Science and Engineering
University of Chittagong

Lecture Overview

- **Probability Theory Review:** Basics of probability, Bayes' theorem.
- **Bayesian Networks:** Representation and inference.
- **Hidden Markov Models (HMMs):** Applications in time-series data.

Lecture Overview

- **Probability Theory Review:** Basics of probability, Bayes' theorem.
- **Bayesian Networks:** Representation and inference.
- **Hidden Markov Models (HMMs):** Applications in time-series data.

Overview of Probability Theory

- **Objective:** Understand randomness in real-world models
- **Key Point:** Probabilistic models deal with uncertainties in phenomena

Sample Space and Events

- **Definition:** The sample space S represents all possible outcomes of an experiment.
- **Example:**

1. If the experiment consists of the flipping of a coin, then

$$S = \{H, T\}$$

where H means that the outcome of the toss is a head and T that it is a tail.

2. If the experiment consists of rolling a die, then the sample space is

$$S = \{1, 2, 3, 4, 5, 6\}$$

where the outcome i means that i appeared on the die, $i = 1, 2, 3, 4, 5, 6$.

Sample Space and Events

- **Example:**

3. If the experiments consists of flipping two coins, then the sample space consists of the following four points:

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

The outcome will be (H, H) if both coins come up heads; it will be (H, T) if the first coin comes up heads and the second comes up tails; it will be (T, H) if the first comes up tails and the second heads; and it will be (T, T) if both coins come up tails.

Sample Space and Events

4. If the experiment consists of rolling two dice, then the sample space consists of the following 36 points:

$$S = \begin{Bmatrix} (1,1), & (1,2), & (1,3), & (1,4), & (1,5), & (1,6) \\ (2,1), & (2,2), & (2,3), & (2,4), & (2,5), & (2,6) \\ (3,1), & (3,2), & (3,3), & (3,4), & (3,5), & (3,6) \\ (4,1), & (4,2), & (4,3), & (4,4), & (4,5), & (4,6) \\ (5,1), & (5,2), & (5,3), & (5,4), & (5,5), & (5,6) \\ (6,1), & (6,2), & (6,3), & (6,4), & (6,5), & (6,6) \end{Bmatrix}$$

where the outcome (i, j) is said to occur if i appears on the first die and j on the second die.

5. If the experiment consists of measuring the lifetime of a car, then the sample space consists of all nonnegative real numbers. That is,

$$S = [0, \infty)^*$$

■

Sample Space and Events

- **Definition:** Any subset E of the sample space S is known as an event.
- **Example:**

1′. In Example (1), if $E = \{H\}$, then E is the event that a head appears on the flip of the coin. Similarly, if $E = \{T\}$, then E would be the event that a tail appears.

2′. In Example (2), if $E = \{1\}$, then E is the event that one appears on the roll of the die. If $E = \{2, 4, 6\}$, then E would be the event that an even number appears on the roll.

3′. In Example (3), if $E = \{(H, H), (H, T)\}$, then E is the event that a head appears on the first coin.

4′. In Example (4), if $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, then E is the event that the sum of the dice equals seven.

5′. In Example (5), if $E = (2, 6)$, then E is the event that the car lasts between two and six years. ■

Union and Intersection of Events

- For any two events **E** and **F** of a sample space **S** we define the new event **E ∪ F** to consist of all outcomes that are either in **E** or in **F** or in both **E** and **F**. That is, the event **E ∪ F** will occur if either **E** or **F** occurs. For example, if **E = {H}** and **F = {T}**, then

$$E \cup F = \{H, T\}$$

- That is, **E ∪ F** would be the whole sample space **S**. If **E = {1, 3, 5}** and **F = {1, 2, 3}**, then

$$E \cup F = \{1, 2, 3, 5\}$$

and thus **E ∪ F** would occur if the outcome of the die is 1 or 2 or 3 or 5. The event **E ∪ F** is often referred to as the union of the event **E** and the event **F**.

Union and Intersection of Events

- For any two events **E** and **F**, we may also define the new event **EF**, sometimes written **E∩F**, and referred to as the intersection of **E** and **F**, as follows. **EF** consists of all outcomes which are both in **E** and in **F.** That is, the event **EF** will occur only if both **E** and **F** occur. For example, if **E = {1, 3, 5}** and **F = {1, 2, 3}**, then

EF = {1, 3}

and thus **EF** would occur if the outcome of the die is either 1 or 3. If **E = {H}** and **F = {T}**, then the event **EF** would not consist of any outcomes and hence could not occur. To give such an event a name, we shall refer to it as the null event and denote it by **Ø**. (That is, **Ø** refers to the event consisting of no outcomes.) If **EF = Ø**, then **E** and **F** are said to be mutually exclusive.

Probabilities Defined on Events

Consider an experiment whose sample space is S. For each event E of the sample space S, we assume that a number $P(E)$ is defined and satisfies the following three conditions:

(i) $0 \leqslant P(E) \leqslant 1$.

(ii) $P(S) = 1$.

(iii) For any sequence of events $E_1, E_2, \ldots$ that are mutually exclusive, that is, events for which $E_n E_m = \emptyset$ when $n \neq m$, then

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n)$$

We refer to $P(E)$ as the probability of the event E.

Probabilities Defined on Events

- **Example 1.**

  In the coin tossing example, if we assume that a head is equally likely to appear as a tail, then we would have

  $$\mathbf{P(\{H\}) = P(\{T\}) = 1/2}$$

Probabilities Defined on Events

- **Example 2.**

In the die tossing example, if we supposed that all six numbers were equally likely to appear, then we would have

P({1}) = P({2}) = P({3}) = P({4}) = P({5}) = P({6}) = 1/6

Moreover, the probability of getting an even number would equal

P({2, 4, 6}) = P({2}) + P({4}) + P({6}) = 1/2

Probabilities Defined on Events

Derive a formula for **P(E ∪ F)**, the probability of all outcomes either in **E** or in **F**.

$$P(E) + P(F) = P(E \cup F) + P(EF)$$

P(E): The probability that event **E** occurs.

P(F): The probability that event **F** occurs.

P(E∪F): The probability that either event **E**, or event **F**, or both occur. This is called the union of **E** and **F**. It represents the event where at least one of **E** or **F** occurs.

P(EF) (or P(E∩F)): The probability that both events **E** and **F** occur. This is called the intersection of **E** and **F**.

Probabilities Defined on Events

Non-overlapping events (Mutually Exclusive):

If **E** and **F** are mutually exclusive (i.e., they cannot happen at the same time, so **P(EF)=0**), then:

$$P(E)+P(F)=P(E \cup F)$$

This makes sense because when events are mutually exclusive, the probability of either event occurring is just the sum of the probabilities of each event, since there's no overlap.

Probabilities Defined on Events

Example 1.3 Suppose that we toss two coins, and suppose that we assume that each of the four outcomes in the sample space

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

Ref: Introduction to Probability Models Tenth Edition, Sheldon M. Ross

is equally likely and hence has probability $\frac{1}{4}$. Let

$$E = \{(H, H), (H, T)\} \quad \text{and} \quad F = \{(H, H), (T, H)\}$$

That is, E is the event that the first coin falls heads, and F is the event that the second coin falls heads.

By Equation (1.2) we have that $P(E \cup F)$, the probability that either the first or the second coin falls heads, is given by

$$\begin{aligned} P(E \cup F) &= P(E) + P(F) - P(EF) \\ &= \tfrac{1}{2} + \tfrac{1}{2} - P(\{H, H\}) \\ &= 1 - \tfrac{1}{4} = \tfrac{3}{4} \end{aligned}$$

This probability could, of course, have been computed directly since

$$P(E \cup F) = P(\{(H, H), (H, T), (T, H)\}) = \tfrac{3}{4}$$ ■

Probabilities Defined on Events

Example: Coin Toss

When you flip a coin, there are two possible outcomes:

- **Event H:** The result is heads.
- **Event T:** The result is tails.

These two events are mutually exclusive because if the coin shows heads, it cannot show tails at the same time, and vice versa. In other words:

- If event **H** occurs (the coin lands on heads), event **T** (the coin landing on tails) cannot occur.
- If event **T** occurs (the coin lands on tails), event **H** cannot occur.

Thus, only one of the two events can happen in any single coin flip, but not both. In terms of probabilities:

- **$P(H \cap T)=0$** because there is no outcome where both heads and tails can happen simultaneously.

Conditional Probabilities

Conditional probability, denoted as **P(A|B)**, represents the probability of an event **A** occurring, given that another event **B** has already occurred.

The conditional probability of A given B, where $P(B) > 0$, is mathematically defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Where:

- $P(A|B)$ is the conditional probability of A happening given that B has occurred.
- $P(A \cap B)$ is the joint probability of both A and B happening.
- $P(B)$ is the probability of B occurring.

Conditional Probabilities

Example: We roll two dice, and we want to find the conditional probability that the sum of the two dice is 9, given that the first die shows a 4.

Step 1: Problem Definition

We are dealing with two events:

Event A: The sum of the two dice equals 9.

Event B: The first die shows 4.

We need to find **P(A|B)**, the probability that the sum of the two dice is 9, given that the first die shows 4.

Conditional Probabilities

Step 2: Total Possible Outcomes

Since each die has 6 faces, the total number of possible outcomes when rolling two dice is: **6×6=36**

So, the sample space S consists of 36 possible outcomes:

$$S=\{(1,1),(1,2),\ldots,(6,6)\}$$

Conditional Probabilities

Step 3: Event A: The Sum is 9

Now, we look at all the possible outcomes where the sum of the two dice equals 9. These outcomes are:
A={(3,6),(4,5),(5,4),(6,3)}

Thus, there are 4 possible outcomes where the sum equals 9. Therefore, the probability of event A (without any additional information) is **P(A)= 4/36 = 1/9**

Step 4: Event B: The First Die is 4

Next, we look at all the possible outcomes where the first die shows a 4. These outcomes are
B={(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)} Thus, there are 6 possible outcomes where the first die is 4. Therefore, the probability of event B is **P(B)= 6/36=1/6**

Conditional Probabilities

Step 5: Event A∩B: The Sum is 9 and the First Die is 4

Now, we find the outcome where both events **A** and **B** occur simultaneously (i.e., the sum is 9 and the first die is 4). For this to happen, the second die must show 5, because:

$$4+5=9$$

Thus, the only outcome where both events **A** and **B** occur is:

$$A \cap B = \{(4,5)\}$$

So, there is 1 outcome where the sum is 9 and the first die is 4. Therefore, the probability of both events occurring is $P(A \cap B) = 1/36$

Conditional Probabilities

Step 6: Calculate the Conditional Probability

Now, we apply the formula for conditional probability:

$$\mathbf{P(A|B)=P(A\cap B)/P(B)}$$

Substitute the known values: $\boldsymbol{P(A|B)=(1/36)/(6/36)=1/6}$ Thus, the conditional probability that the sum of the two dice is 9, given that the first die shows 4, is: $\boldsymbol{P(A|B)=1/6}$

Initially, the probability of rolling a sum of 9 was 1/9. However, once you know that the first die is a 4, the number of possible outcomes is reduced from 36 to just 6 (the cases where the first die is 4). Out of those 6 possible outcomes, only 1 gives a sum of 9 (when the second die is 5). Therefore, the conditional probability is 1/6 .This shows how knowing additional information (that the first die shows 4) changes the probability of rolling a sum of 9.

Conditional Probabilities

Example: Suppose cards numbered one through ten are placed in a hat, mixed up, and then one of the cards is drawn. If we are told that the number on the drawn card is at least five, then what is the conditional probability that it is ten?

We need to calculate the conditional probability that the card drawn is **10**, given that the number on the card is at least **5**.

Step 1: Define the Events

Let:

- **A** be the event that the card drawn is **10**.
- **B** be the event that the card drawn is at least **5**.

We are looking for the conditional probability **P(A|B)**, which is the probability that the card is **10** given that the card drawn is at least **5**.

Conditional Probabilities

Step 2: Apply the Formula for Conditional Probability

The formula for conditional probability is:

$$P(A|B) = P(B)P(A \cap B)$$

Where: **P(A∩B)** is the probability that the card is both **10** and at least **5**. **P(B)** is the probability that the card is at least **5**.

Conditional Probabilities

Step 3: Calculate P(A∩B)

Since **A** is the event that the card is **10**, and **10** is clearly at least **5**, the event **A∩B** is just the event that the card is **10.** The probability of drawing the card numbered **10** is: $P(A \cap B) = P(\text{card is 10}) = 1/10$

Conditional Probabilities

Step 4: Calculate P(B)

The event **B** is that the card drawn is at least **5**. The cards numbered at least **5** are **{5,6,7,8,9,10}**. There are **6** possible cards that satisfy this condition, so the probability of drawing a card that is at least **5** is:

$P(B)$**=6/10=0.6**

Conditional Probabilities

Step 5: Calculate the Conditional Probability

Now, using the conditional probability formula:
$P(A|B)=P(A \cap B)P(B)=(1/10)/(6/10)=1/6$

The conditional probability that the card is **10**, given that the card is at least **5**, is: $P(A|B)=1/6$

Bayesian Classification (ML Method for Supervised Learning)

- **A statistical classifier:** performs probabilistic prediction, i.e., predicts class membership probabilities
- **Foundation:** Based on Bayes' Theorem.
- **Performance:** A simple Bayesian classifier, naïve Bayesian classifier, has comparable performance with decision tree and selected neural network classifiers
- **Incremental:** Each training example can incrementally increase/decrease the probability that a hypothesis is correct — prior knowledge can be combined with observed data
- **Standard:** Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which other methods can be measured.

Bayes' Theorem: Basics

- **Total probability Theorem:** $P(B) = \sum_{i=1}^{M} P(B|A_i)P(A_i)$
- **Bayes' Theorem:** $P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})} = P(\mathbf{X}|H) \times P(H) / P(\mathbf{X})$
  - Let X be a data sample ("evidence"): class label is unknown
  - Let H be a hypothesis that X belongs to class C
  - Classification is to determine P(H|X), (i.e., posteriori probability): the probability that the hypothesis holds given the observed data sample X
  - P(H) (prior probability): the initial probability
  - E.g., X will buy computer, regardless of age, income, …
  - P(X): probability that sample data is observed
  - P(X|H) (likelihood): the probability of observing the sample X, given that the hypothesis holds
  - E.g., Given that X will buy computer, the prob. that X is 31..40, medium income

Bayes Theorem Derivation

Derivation of Bayes' Theorem:

Bayes' Theorem is derived from the definition of **conditional probability**. By definition:

$$P(H|E) = \frac{P(H \cap E)}{P(E)}$$

This represents the probability of H and E occurring together, normalized by the probability of the evidence E.

Similarly, the joint probability $P(H \cap E)$ can also be expressed as:

$$P(H \cap E) = P(E|H) \cdot P(H)$$

This gives us the **joint probability** of both H and E occurring. Substituting this into the equation for conditional probability, we get:

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Bayes Theorem Derivation

Law of Marginal Likelihood:

The denominator in Bayes' Theorem, $P(E)$, is called the **marginal likelihood** or the **evidence**. It represents the overall probability of observing the evidence E across all possible hypotheses. If we have several mutually exclusive and exhaustive hypotheses $H_1, H_2, \ldots, H_n$, the marginal likelihood is calculated using the **law of total probability**:

$$P(E) = \sum_{i=1}^{n} P(E|H_i) \cdot P(H_i)$$

In other words, $P(E)$ is the weighted sum of the likelihoods of E under all possible hypotheses, where each likelihood is weighted by the prior probability of the corresponding hypothesis.

Prediction Based on Bayes' Theorem

- Given training data **X**, posteriori probability of a hypothesis H, P(H|X), follows the Bayes' theorem

$$P(H|\mathbf{X}) = \frac{P(\mathbf{X}|H)P(H)}{P(\mathbf{X})} = P(\mathbf{X}|H) \times P(H) / P(\mathbf{X})$$

- Informally, this can be viewed as

  posteriori = likelihood x prior/evidence

- Predicts **X** belongs to C_i iff the probability $P(C_i|\mathbf{X})$ is the highest among all the $P(C_k|\mathbf{X})$ for all the k classes
- Practical difficulty: It requires initial knowledge of many probabilities, involving significant computational cost

Classification Is to Derive the Maximum Posteriori

- Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n-D attribute vector **X** = (x_1, x_2, …, x_n)
- Suppose there are *m* classes C_1, C_2, …, C_m.
- Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|\mathbf{X})$
- This can be derived from Bayes' theorem

$$P(C_i|\mathbf{X}) = \frac{P(\mathbf{X}|C_i)P(C_i)}{P(\mathbf{X})}$$

- Since P(X) is constant for all classes, only

$$P(C_i|\mathbf{X}) = P(\mathbf{X}|C_i)P(C_i)$$

needs to be maximized

Naïve Bayes Classifier

- A simplified assumption: attributes are conditionally independent (i.e., no dependence relation between attributes):

$$P(\mathbf{X}|C_i) = \prod_{k=1}^{n} P(x_k|C_i) = P(x_1|C_i) \times P(x_2|C_i) \times ... \times P(x_n|C_i)$$

- This greatly reduces the computation cost: Only counts the class distribution

Naïve Bayes Classifier: Training Dataset

Class:
C1:buys_computer = ‘yes’
C2:buys_computer = ‘no’

Data to be classified:
X = (age <=30,
Income = medium,
Student = yes
Credit_rating = Fair)

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier: An Example

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

- $P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643
  P(buys_computer = "no") = 5/14= 0.357
- Compute $P(X|C_i)$ for each class
  P(age = "<=30" | buys_computer = "yes") = 2/9 = 0.222
  P(age = "<= 30" | buys_computer = "no") = 3/5 = 0.6
  P(income = "medium" | buys_computer = "yes") = 4/9 = 0.444
  P(income = "medium" | buys_computer = "no") = 2/5 = 0.4
  P(student = "yes" | buys_computer = "yes) = 6/9 = 0.667
  P(student = "yes" | buys_computer = "no") = 1/5 = 0.2
  P(credit_rating = "fair" | buys_computer = "yes") = 6/9 = 0.667
  P(credit_rating = "fair" | buys_computer = "no") = 2/5 = 0.4
- **X = (age <= 30 , income = medium, student = yes, credit_rating = fair)**

$P(X|C_i)$: P(X|buys_computer = "yes") = 0.222 x 0.444 x 0.667 x 0.667 = 0.044
P(X|buys_computer = "no") = 0.6 x 0.4 x 0.2 x 0.4 = 0.019

$P(X|C_i)*P(C_i)$: P(X|buys_computer = "yes") * P(buys_computer = "yes") = 0.028
P(X|buys_computer = "no") * P(buys_computer = "no") = 0.007

Therefore, X belongs to class ("buys_computer = yes")

Naïve Bayes Classifier

Using the Laplacian correction to avoid computing probability values of zero. Suppose that for the class *buys_computer* = *yes* in some training database, *D*, containing 1000 tuples, we have 0 tuples with *income* = *low*, 990 tuples with *income* = *medium*, and 10 tuples with *income* = *high*. The probabilities of these events, without the Laplacian correction, are 0, 0.990 (from 990/1000), and 0.010 (from 10/1000), respectively. Using the Laplacian correction for the three quantities, we pretend that we have 1 more tuple for each income-value pair. In this way, we instead obtain the following probabilities (rounded up to three decimal places):

$$\frac{1}{1003} = 0.001, \frac{991}{1003} = 0.988, \text{ and } \frac{11}{1003} = 0.011,$$

respectively. The "corrected" probability estimates are close to their "uncorrected" counterparts, yet the zero probability value is avoided. ■

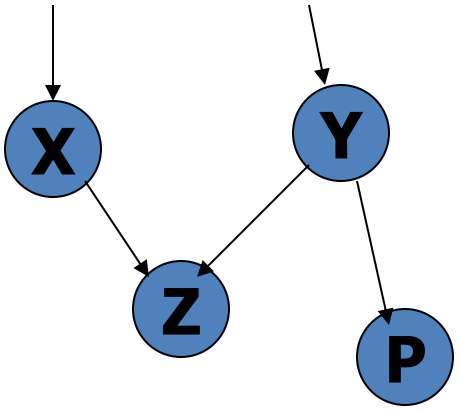
Ref: Data Mining Concepts and Techniques (pdf)
Example 8.5

Lecture Overview

- **Probability Theory Review:** Basics of probability, Bayes' theorem.
- **Bayesian Networks:** Representation and inference.
- **Hidden Markov Models (HMMs):** Applications in time-series data.

Bayesian Network

- **Bayesian belief networks** (also known as **Bayesian networks**, **probabilistic networks**): allow *class conditional independencies* between *subsets* of variables
- A (*directed acyclic*) graphical model of causal relationships
  - Represents dependency among the variables
  - Gives a specification of joint probability distribution



- ❑ Nodes: random variables
- ❑ Links: dependency
- ❑ X and Y are the parents of Z, and Y is the parent of P
- ❑ No dependency between Z and P
- ❑ Has no loops/cycles

Joint Probability Distribution

The joint probability distribution of a Bayesian network is the product of the conditional probabilities of each node given its parents in the network. This means that the joint probability distribution provides a complete description of the probability distribution of all the variables in the network.

Mathematically, the joint probability distribution of a Bayesian network can be represented as:

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i | parent(X_i))$$

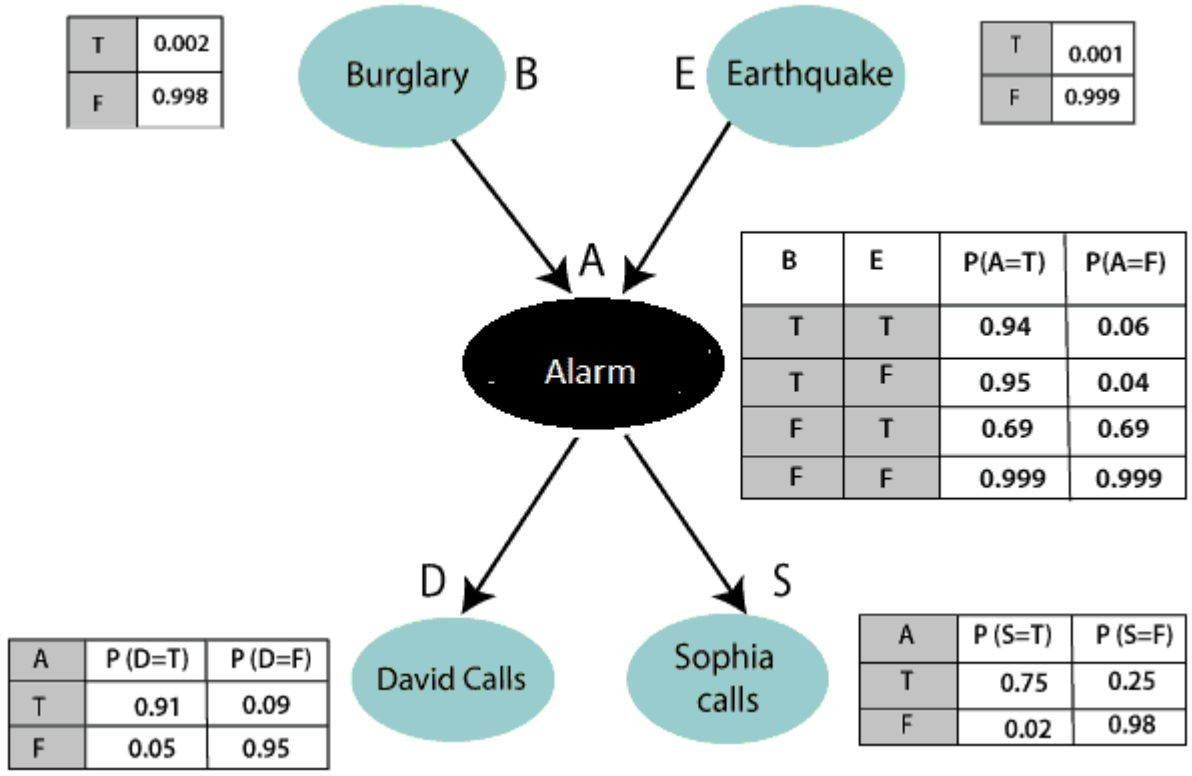
Where $X_1, X_2, \ldots, X_n$ are random variables in the network, and $parent(X_i)$ represents the set of parent nodes of X_i.

Explanation of Bayesian Network

Let’s create a Bayesian Belief Network (BBN) based on a classic example involving an alarm system, burglary, and an earthquake. In this scenario, we will model the relationships between these events, including how they affect each other and how we can infer probabilities from them.

Explanation of Bayesian Network

Problem: Calculate the probability that alarm has sounded, but there is neither a burglary, nor an earthquake occurred, and David and Sophia both called the Neighbor.



T	0.002
F	0.998

T	0.001
F	0.999

B	E	P(A=T)	P(A=F)
T	T	0.94	0.06
T	F	0.95	0.04
F	T	0.69	0.69
F	F	0.999	0.999

A	P (D=T)	P (D=F)
T	0.91	0.09
F	0.05	0.95

A	P (S=T)	P (S=F)
T	0.75	0.25
F	0.02	0.98

Explanation of Bayesian Network

Solution:

- The network topology shows that burglary and earthquake are the parent nodes of the alarm, affecting the probability of the alarm going off directly, but David and Sophia's calls are dependent on the probability of the alarm going off.
- Conditional probabilities tables, or CPTs, are used to represent the conditional distributions for each node.
- Because all of the entries in the table indicate an exhaustive set of cases for the variable, each row in the CPT must add to 1.
- 2^k probabilities are included in a boolean variable with k boolean parents in CPT. As a result, if there are two parents, CPT will have four probability values.

Explanation of Bayesian Network

Solution:

- We may formulate the issue statement in the form of a probability distribution using the joint distribution formula:

$$P(S, D, A, \neg B, \neg E) = P(S|A) * P(D|A) * P(A|\neg B \wedge \neg E) * P(\neg B) * P(\neg E).$$

$$= 0.75 * 0.91 * 0.001 * 0.998 * 0.999$$

$$= 0.00068045.$$

Lecture Overview

- **Probability Theory Review:** Basics of probability, Bayes' theorem.
- **Bayesian Networks:** Representation and inference.
- **Hidden Markov Models (HMMs): Applications in time-series data.**

Hidden Markov Models (HMMs)

- Hidden Markov Models (HMMs) are probabilistic models used to represent systems that are assumed to follow a Markov process with hidden (unobservable) states.
- Two key components:
  - Hidden States: Actual states that the system is in but cannot be directly observed.
  - Observations: Observable outputs influenced by the hidden states.

Three Key Problems in HMMs

- **Evaluation Problem (Likelihood):** Given an HMM λ and a sequence of observations O=$o1$,$o2$,…,oT, calculate the probability $P(O|\lambda)$.
- **Decoding Problem (State Sequence):** Given an observation sequence, find the most likely hidden state sequence that produced the observations.
- **Learning Problem (Parameter Estimation):** Adjust the HMM parameters to maximize the likelihood of the observation sequence.

Evaluation Problem: Forward Algorithm

- The **forward algorithm** is used to compute $P(O|\lambda)$ efficiently.
- **Steps:**

1. Initialize $\alpha_1(i) = \pi_i b_i(o_1)$
2. Recursively compute $\alpha_t(j) = \sum_{i=1}^{N} \alpha_{t-1}(i) a_{ij} b_j(o_t)$
3. Compute total probability $P(O|\lambda) = \sum_{i=1}^{N} \alpha_T(i)$

Applications of Hidden Markov Models

- **Speech Recognition:** HMMs are used to model phonemes as hidden states and speech features as observations.
- **Natural Language Processing:** Part-of-speech tagging and named entity recognition.
- **Bioinformatics:** Gene prediction and sequence alignment.
- **Time-Series Analysis:** Stock market prediction and weather forecasting.

HMMs Conclusion

- Hidden Markov Models provide a powerful framework for modeling sequential data where the underlying process is hidden.
- HMMs are widely used in areas such as speech recognition, NLP, and bioinformatics.
- Understanding the three main problems in HMMs — **evaluation, decoding, and learning** — is key to applying them effectively.

Book Reference and Reading Materials

- **Stuart Russell and Peter Norvig: "Artificial Intelligence a Modern Approach", 3rd Edition, Pearson Education Asia. (Text Book) (pdf) (Lectures) Chapter 13,14,15.**

- **Introduction to Probability Models ,Tenth Edition**
  Sheldon M. Ross
  University of Southern California
  Los Angeles, California

- **Data Mining Concepts and Techniques (pdf)**
  Third Edition
  Jiawei Han
  University of Illinois at Urbana–Champaign
  Micheline Kamber
  Jian Pei
  Simon Fraser University

- **Bayesian Inference for AI.pdf (In Lecture folder)**